

AI Ops Assignment 1

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Question 1

Part A

The major technical debt response for this is that of **Entanglement**. Since the changing of the estimated delivery time hurt another unrelated feature we can say that it is because they are entangled and leading to "Changing Anything Changes Everything".

Part B

The major technical debt causing this is **Undeclared Consumers**. Since the other teams were not aware that the marketing team was reading the raw output from the table, it is an example of consumers who are undeclared resulting in undesirable consequences.

Part C

The major technical debt causing this is **Configuration and Glue Code Debt** since the exact commands are not being tracked and are scattered in no specific place, it makes the code harder to keep a track of.

Part 2: Fixing Part C

To fix the issues arising from part C we can make use of tracking and versioning the code to prevent the parameters from being scattered. Additionally we can add an orchestration tool for the model by packaging the model using mlflow and then the model can be run with a single line command.

Question 2

Part 2

The best performing run appears to be the run with alpha as 0.01, learning rate as 0.001 and batch size as 64. This is likely because increasing the learning rate beyond this point leads to the model beginning to have updates which are too large and it becomes unable to precisely find the minima point. A larger batch size is likely also better for the same reason as its updates contain less noise. For this specific set of runs we observe no overfitting occurring and that is likely because the variation in the hyperparameters is small, if we were to vary the hyperparameters between larger ranges we would likely see the model begin to overfit especially for a larger number of epochs. We seem to get a noticeable improvement from the batch size hyper parameter as changing it from 32 to 64 seems to raise the accuracy and make the model have better more precise updates. To get better estimates of which hyperparameters are accurate we can run grid search over a larger set of hyperparameters and precisely identify what are the largest contributing factors.

Part 3

Part 3 can be found as a python file of the logging snippet as well as the entire ipynb used for the video have both been placed in the github repository.